pccx: An Open, Sail-Specified W4A8 NPU for Edge LLM Inference on the Xilinx Kria KV260

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Abstract

We present **pccx**, an open, scalable Neural Processing Unit (NPU) architecture for autoregressive decoding of Transformer-based large language models on resource-constrained edge devices. The v002 reference implementation targets the Xilinx Kria KV260 (Zynq UltraScale+ ZU5EV), is formally specified in the Sail [3] ISA semantics language (the same language used for RISC-V, Arm, CHERI and Morello), and implements recent W4A8KV4 quantisation [5], FlashAttention [4], and NPU-adapted speculative decoding [2] in a single 64-bit VLIW ISA with five base opcodes (GEMV / GEMM / MEMCPY / MEMSET / CVO). An accompanying desktop profiler, **pccx-lab**, supplies 16 built-in trace analyzers and an AI-driven UVM strategy copilot. This preprint covers the architecture, the Sail model, the profiler design, and the current measurement status.

1 Introduction

TODO — three paragraphs:

- Edge LLM inference landscape (Gemma 3N, Llama, Phi).
- Why W4A8KV4 on FPGA makes sense for a KV260-class device.
- pccx contribution summary; pointer to the three repos at `https://github.com/hwkim-dev/`.

2 Architecture

2.1 Compute cores

32×32 systolic GEMM array with cascade break at row 16; 4-lane × 32-MAC GEMV with 5-stage reduction; single BF16 SFU pipeline covering EXP / SQRT / GELU / SIN / COS / REDUCE_SUM / SCALE / RECIP. Weight stationary in GEMM, weight streaming in GEMV.

2.2 Memory hierarchy

1.75 MB L2 URAM at the geometric centre of the floorplan. Dual-port access: ACP for latency-critical host path, NPU port-B for direct compute fills. Constant cache with 64 entries, 6-bit pointer from the ISA. HP0/1 carry GEMM weights, HP2/3 carry GEMV weights; both pairs independent at 128-bit × 250 MHz.

2.3 ISA

64-bit VLIW with a 4-bit opcode in bits `[63:60]` and a 60-bit body. Five opcodes:

Opcode	Encoding	Body layout
`GEMV`	`0x0`	dst(17) ‖ src(17) ‖ flags(6) ‖ size(6) ‖ shape(6) ‖ lanes(5) ‖ rsv(3)
`GEMM`	`0x1`	identical to `GEMV`
`MEMCPY`	`0x2`	from(1) ‖ to(1) ‖ dst(17) ‖ src(17) ‖ aux(17) ‖ shape(6) ‖ async(1)
`MEMSET`	`0x3`	cache(2) ‖ dst(6) ‖ a(16) ‖ b(16) ‖ c(16) ‖ rsv(4)
`CVO`	`0x4`	func(4) ‖ src(17) ‖ dst(17) ‖ len(16) ‖ flags(5) ‖ async(1)

3 Sail Formal Specification

The ISA is authored in **Sail** [3], with modules split incrementally per the language conventions:

```
formal/sail/
  pccx.sail_project
  src/prelude.sail # base bitvectors
  src/pccx_types.sail # opcode table, body structs, flags
  src/pccx_regs.sail # cycle / pc / committed_any
  src/pccx_decode.sail # 64-bit word -> typed instr union
  src/pccx_execute.sail # step(); advance_cycle; record_event()
  tests/smoke_decode.sail
```

Every SystemVerilog `typedef` in the RTL ISA package has a 1:1 Sail counterpart, so width drift fails the Sail type checker before the synthesiser.

4 pccx-lab Profiler

TODO — describe:

- 16 built-in `TraceAnalyzer` implementations.
- `Copilot::investigate` / `suggest_fix` surface.
- Research-citation registry (21 papers auto-cited).
- Tauri desktop UI with WaveformViewer, FlameGraph, ISA replay.

5 Methodology

Two sample traces are bundled with the repository (`samples/`, 16- and 128-token decode) so every number in this paper is reproducible. See the Quickstart at https://hwkim-dev.github.io/pccx/en/docs/quickstart.html.

6 Evidence

TODO — fill as measurements land:

- Gemma-3N E4B tokens per second on KV260.
- KV260 resource usage (LUT / DSP / URAM / BRAM36).
- Timing closure @ 400 MHz core / 250 MHz AXI.
- CPU baseline (Ryzen 4500U, llama.cpp Q4_K_M).

- GPU baseline (RTX 4060, HF Transformers BF16).

Current status is tracked live at https://hwkim-dev.github.io/pccx/en/docs/Evidence/index.html.

7 Related Work

Sail precedents for ISA formalisation: [3] for RISC-V; [7] for Armv8-A. W4A8KV4 lineage: [5] (QServe), [8] (QQQ), [1] (SRAM–frequency bottleneck model). FlashAttention: [4] (FA-2), [6] (FA-3). Speculative decoding on static-graph NPUs: [2].

8 Conclusion and Future Work

TODO — one paragraph: the value of combining open RTL + formal Sail spec + profiler; near-term goal of board-captured tokens per second; longer-term goals (Gemma-4, Isabelle proofs).

References

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[6] Jay Shah, Ganesh Bikshandi, Ying Zhang, and Tri Dao. FlashAttention-3: Fast and accurate attention with asynchrony and low-precision. *arXiv preprint arXiv:2407.08608*, 2024.

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